

HS4002: Lecture 3A

Introduction to Surveys

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What is a Survey?

A Broad Definition

- The survey is a research method where we try to collect responses from a group of individuals.
 - It's primarily viewed as a method of data collection, but sometimes people conflate as a way of sampling.
- Three things to think about:
 1. How do we collect these responses?
 2. What kinds of responses are we looking for?
 3. How do we choose this group of individuals?

How Are Surveys Conducted?

- Surveys can be conducted in many ways.
- Four major modes
 - In-person
 - Mail
 - Telephone
 - Web/internet
- We can also do mixed-mode surveys:
 - Concurrent mixed-mode surveys
 - Sequential mixed-mode surveys

What Do Surveys Ask?

- Usually, respondents are given a set of pre-determined choices.
- Sometimes, open-ended.

Who Gets Surveyed?

- Identify a population, and then an appropriate sampling frame.
- Probability Sampling:
 - Identify a population, and then an appropriate sampling frame.
 - Stratify if needed.
 - Sample.
- Nonprobability sampling:
 - Convenience sampling
 - Snowball sampling
 - Respondent-driven sampling
 - Open-access Surveys

Why Do Probability Sampling?

- This is related to the statistical theory you learnt previously.
- We know how to produce an unbiased estimate of important statistics from just a sample from the population.
- The statistical distribution of descriptive statistics from a probability sample of finite populations has been well-known since the 1930s.
- This ability to get good estimates from samples is conditional on the fact that any observed variation in the sample is RANDOM, and not generated by some unobserved process.
- We can save ourselves the time and resources required to do a full census of the population.

How Have Surveys Changed?

- Began with in-person surveys to sampled addresses.
- Telephone became the dominant mode of surveys in the late 20th century.
 - Steep decline in response rates: from 36% in 1997 to 9% in 2012
 - Tele-marketing is often blamed for this.
- Online data collection has become the new frontier.

Classroom Exercise I:

With the past Singapore Parliamentary Election in mind, I'd like you to compare an ideal-typical probability survey to a national election. To what extent are they similar? To what extent are they different?

Back to Week 2 lab

Step 1: Commit and Push

```
git add week2/week2_notebook_student.ipynb  
git commit -m "ez"  
git push origin week2
```

Pushes your branch to *your fork* (origin) — not upstream. You can't push there.

Step 2: Open the Pull Request

- On GitHub, go to your fork — a banner offers “Compare & pull request” for the branch you just pushed. Click it.
- Confirm:
 - **base repository:** the class repo, base main
 - **head repository:** your fork, compare your branch
- Fill in the PR template (name, matric number, week) and submit.

Evaluating Surveys

- Total survey error (TSE) is a framework for thinking about all the sources of errors that might affect the statistical properties of our survey estimates.
- TSE is the difference between an estimate produced by the survey data, and the true value of the estimand in the population.
- Surveys are better when they have lower TSE.

Four Sources of TSE

1. Coverage Error
2. Sampling Error
3. Nonresponse Error
4. Measurement Error

- Errors produced from our choice of sampling frame.
- Example:
 - Landline random digit dialing may be a biased sampling frame today even though it wasn't in 1993.
 - A university student experience survey that is distributed through student dormitories.

- Errors that are inherent to the individuals sampled.
- This existed in all surveys, and can only be ameliorated by sample size.
- Cannot be removed; only needs to be reduced to acceptable levels.

Nonresponse Error

- It is not just response rate that matters; it's who chooses to respond.
- Errors that are a result of a systematic divergence between those who respond to a survey, and those who do not respond to a survey.
- Example:
 - Internet open-access reviews.

- Errors that are caused by the construction of the survey item/response.
- Construct validity (whether an indicator is a good representation for a given concept) is related.
- Two main types
 - Response bias: systematic variation in measurement error.
 - Response variance: poor reliability, where measurement upon re-measures.

Classroom Exercise II

Use the TSE to identify sources of errors in this research design:

1. I'm interested in Singaporeans' music tastes.
2. I take a random sample of 500 NTU student emails.
3. I ask them two questions in the survey.
4. "Now I am going ask about your music preferences. For each type of music I read, please tell me if you like to listen to it. Do you like to listen to (hip-hop/k-pop/mando-pop)?"
→ Yes/No response options
5. "Following up, how many times last week did you listen to (hip-hop/k-pop/mando-pop) music?"
→ Open-ended box

What Might Not Be Considered By TSE?

- Is a item/measure credible?
- Is a survey commensurable or comparable to others?

Encouraging Survey Responses

How Can We Improve Response Rates

Dillman et al. (2012) suggest we think in terms of (a) benefits, (b) costs, and (c) trust.

Increase the Benefits of Participation.

- Make it clear how a survey is helpful or useful.
- Ask interesting questions
- Borrow legitimacy.
- Use material incentives where appropriate.

Decrease the Costs of Participation.

- Reduce cognitive load: shorten and simplify questions as best as you can.
- Adopt good visual principles
- Make response convenient
- Allow alternative modes of responses

- Make it possible for respondents to verify authenticity of survey request
- Borrow legitimacy from associated organizations.
- Assure confidentiality.
- Look professional.

TODOS

TODO for next week

Scoping memo is due next week (Wednesday, 10pm). And start preparing for your group facilitation weeks (for certain groups).

Thanks for listening :)